

Shivansh

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Education

Delhi Technological University (DTU), Delhi

Expected May 2028

B.Tech in Software Engineering

GPA: 7.4/10

Relevant Coursework: Data Structures & Algorithms, Machine Learning, Database Systems, Software Engineering, Computer Networks

Work Experience

ONGC, India

Dec 2025 – Jan 2026

Software Intern – Rig Power Systems & BESS Feasibility

- Cleaned and performed EDA on rig power datasets (**10,000+ records**) using Python (Pandas, NumPy); resolved sensor noise, outliers, and missing values across multiple drilling sites.
- Identified generator over-utilization and load imbalance patterns, flagging **15–20% fuel efficiency gaps**; delivered a BESS feasibility report adopted by the engineering team.

IIT Mandi

Dec 2025 – Present

Research Intern – EEG Signal Modeling (Self-Supervised Learning)

- Designed a Transformer + CBAM attention SSL model for EEG representation learning with contrastive pipelines (time-masking, channel-dropout, phase-shuffling augmentations).
- Achieved strong low-label performance (<10% labeled data), reducing dependency on costly annotated EEG recordings across clinical classification tasks.

Projects

SmartCourt – AI Football Tactical Analysis

[GitHub]

PyTorch, GNN, GAT, CVAE, StatsBomb 360

- Built a GNN + Graph Attention Transformer to model player interactions from StatsBomb 360 data; implemented CVAE for counterfactual defensive scenario simulation.
- Engineered spatial-temporal graph features from raw event streams; end-to-end pipeline processes a full match in under 3 minutes.

DABOT – AI DSA Tutor Chatbot

[GitHub]

Python, OpenAI, QWEN, RAG, FastAPI

- Built a multimodal AI chatbot for DSA tutoring with step-by-step explanations; integrated RAG over a curated knowledge base to improve factual accuracy and reduce hallucinations.
- Deployed as a FastAPI web application supporting real-time query resolution with <2s average response latency.

Water Quality Analysis – MahaKumbh 2025

[GitHub]

- Evaluated water quality at **7 Ganga-Yamuna river sites** using CCME WQI 1.0; performed statistical analysis and time-series visualization across multi-parameter environmental datasets.
- Identified marginal quality bands (WQI 49–63) and seasonal pollution spikes; findings compiled into a policy-level environmental report.

Technical Skills

Languages: Python (primary), C++, SQL

ML / DL: PyTorch, Scikit-learn, GNN, GAT, CVAE, Transformers, CBAM, SSL (contrastive learning)

Data & Vision: NumPy, Pandas, OpenCV, Matplotlib, Tableau

Tools: Git, Linux, FastAPI

Domains: Machine Learning, Deep Learning, Computer Vision, NLP, Graph Neural Networks, RAG

Leadership & Activities

Co-Head, Paryavarnam – DTU Environmental Society

Sep 2025 – Present

- Led a 20-member core team to organize DTU's annual sustainability drive with **2,000+ participants** across 3 colleges; initiated a campus waste-audit project producing a report submitted to administration.

Achievements

- **Finalist**, TechFest MAIT (2025) – SmartCourt AI project selected among top teams across Delhi NCR.
- Conducted AI/ML workshops for **200+ students** (Classes 10–12) covering tools, career paths, and practical demos.
- Secured **2nd place**, Spiritual Play Competition, TOVP Mayapur (July 2024).